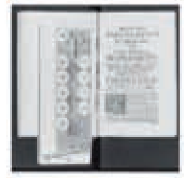
Computers

Computers are electronic machines that we can use to do many different things, just by changing the programs they are running. Today, computers have become indispensable because they are used to run our world – from global air traffic control to personal mobile phones.

c.2000 BCE
The Chinese invent the abacus, the world's first counting machine.

Abacus



1666
Samuel Morland invents a machine that can add and subtract.

Morland's calculating machine

2000 BCE

COMPUTER HISTORY

The first calculating machines were invented to add numbers, which was important for buying and selling goods. They were continually improved, until we arrived at the modern computer.

1642
Blaise Pascal invents the Pascaline, a mechanical and automatic calculator.

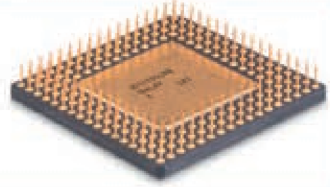
Pascaline

HOW COMPUTERS WORK

Computers work by processing information: they take in information (data), store it (memory), process it in whichever way they have been programmed to do, then display the result (output).

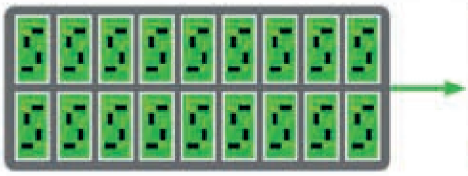
PROCESSOR CHIP

A computer's processor is like the brain of the computer. It uses a chip – a piece of silicon that can hold billions of components – to perform its computing tasks.



KEYBOARD
The keyboard and mouse are both input units – they are ways of getting information into the computer.

MOUSE



MEMORY
Computers use two different kinds of memory: ROM (read-only memory), and RAM (memory you can change).

SCREEN
Computers have LCD screens to display the result of the processing that has taken place.

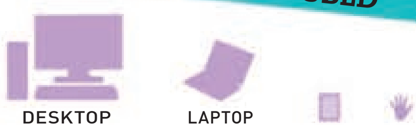
INPUT | STORAGE AND PROCESSING | OUTPUT

SHRINKING SIZES

The 1949 EDSAC computer took up a whole room and was arranged over 12 racks. Today's personal computers (PCs) perform calculations millions of times faster, but they can sit easily on someone's desk or lap.

EARLY COMPUTER
1949

THE WORLD'S SMALLEST COMPUTER IS JUST ONE MILLIMETRE CUBED



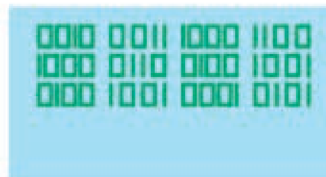
DESKTOP COMPUTER
1980

LAPTOP COMPUTER
1983

TABLET
1993

SOFTWARE

Software is the name for ready-made programs we use to make one computer do many things. Software allows us to write, edit photos, use the internet, and so on, without having to program a computer ourselves.

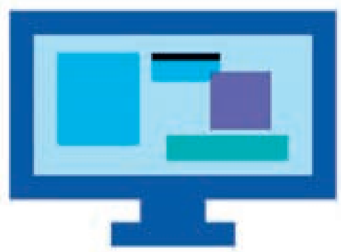


1 BINARY CODE
Computers only understand binary code, which is made up of 0s and 1s.

```
IDLE  File  Edit  Shell  Debug  Window  Help

# Ghost Game
from random import randint
print ( 'Ghost Game' )
feeling_brave=True
Score=0
While feeling_brave:
      ghost_door=randint( 1,3 )
      print('Three doors ahead...')
```

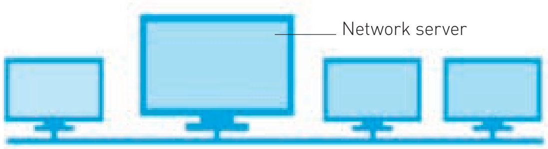
2 PROGRAMMING LANGUAGE
Programming languages are used to lay out sets of instructions for computers to follow.



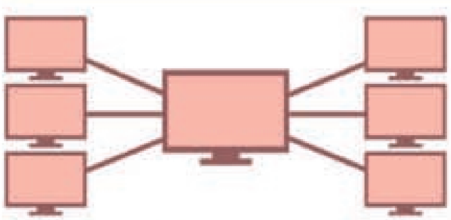
3 SOFTWARE
Programming languages are used to write computer programs (software).

NETWORKS

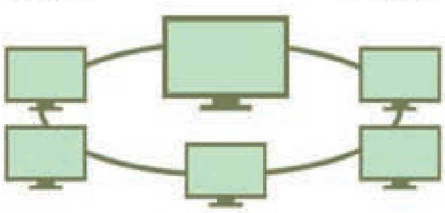
A network is a number of things connected in some way. There are three main forms of computer networks, which can connect computers and peripherals, such as printers.



BUS NETWORK
This simple network uses one cable to connect all the devices in a line. One computer acts as the server.



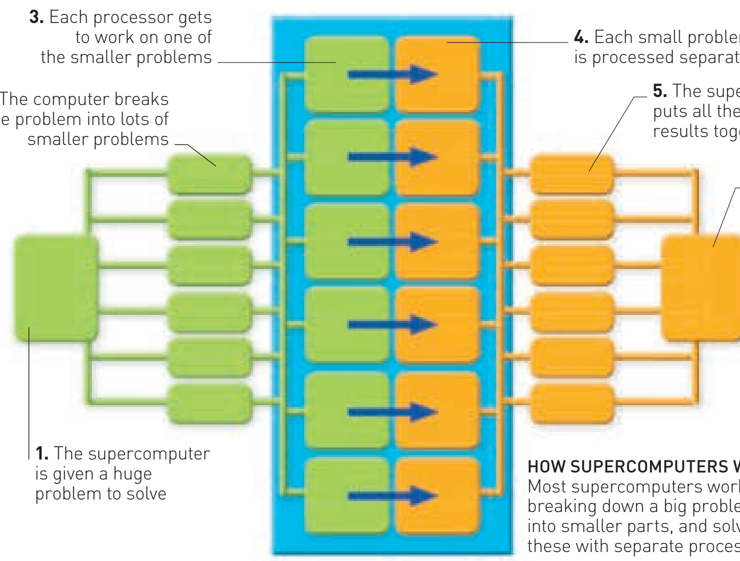
STAR NETWORK
Here, each computer is connected to a central hub, using an individual cable for each computer. If one device fails, the others are not affected.



RING NETWORK
Here, the computers are connected in a circular pattern, and the information travels in one direction. Each computer is connected to the next one.

SUPERCOMPUTERS

Some scientific problems are so vast that they need huge amounts of processing power, delivered by "supercomputers". Some of these have tens of thousands of processors all working on one thing at the same time.



HOW SUPERCOMPUTERS WORK
Most supercomputers work by breaking down a big problem into smaller parts, and solving these with separate processors.